

The Idea, End to End



Energy Grid Heat-Demand Forecaster

In one line: Software that watches the weather and warns the power grid before it gets overwhelmed by everyone turning on their AC at once.



Objective

Give power utilities enough advance warning about heat-driven electricity demand that they can act **before** things go wrong — not after.

Turn FortyGuard's hyperlocal (city-block-level) temperature data into: *"Zone 4 will spike in the next few hours — here's how much, and how long you have to prepare."*



Problem It Solves

- **Heat drives AC use** — demand isn't spread evenly across a city.
- **Grids have limits** — sudden demand forces emergency power or blackouts.
- **Today's forecasting is too broad** — city-wide, not district-by-district.
- **Reaction beats prevention** — by the time demand rises, the calm window to act is gone.



How It Works

- 1 **Heat data**
FortyGuard current + 12hr forecast per zone
- 2 **Cooling Degree Days**
proven utility formula, not a guess
- 3 **Calibrate**
tune with public demand datasets
- 4 **Risk score**
0-100 per-zone demand-spike score
- 5 **Agent loop**
observe→evaluate→plan→validate→act→record, human approves



Why We Stand Out

- **Named vertical** — Energy Trading is one of FortyGuard's 3 named 2026 growth focuses.
- **Real science** — built on a documented, industry-proven formula.
- **Hard to fake** — most teams won't attempt a real model.
- **Boardroom stakes** — blackout prevention gets judge attention.
- **Team fit** — we have the AI depth to build the real version.

Why it's not just a chart: A dashboard displays numbers. Our system continuously watches, reasons, and recommends action to a human — that's what makes it an AI *agent*, matching the hackathon's primary track.

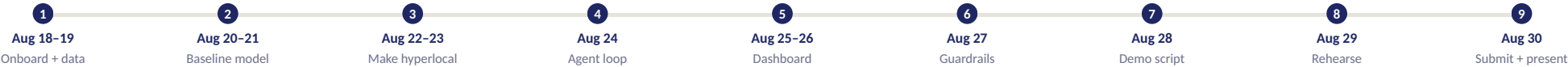
The Plan, and Why It Wins

Build Window: Aug 18–30, 2026



The 13-Day Sprint

From onboarding to submission — every stage mapped to a concrete deliverable.



Team Split

- AI / Modeling Lead**
Cooling-degree-days model, calibration, decision logic
- Data & API Integration**
FortyGuard onboarding, zone pipeline, dataset prep
- Frontend & Dashboard**
Map view, risk cards, timeline, approval UI
- Agent Logic & Guardrails**
Loop, thresholds, audit logging

Why This Is the Best Idea

- ✓ **Named FortyGuard vertical** — one of only 3 named 2026 growth areas.
- ✓ **Real, defensible science** — not an invented formula.
- ✓ **Genuinely hard to fake** — differentiates us from easy dashboard entries.
- ✓ **High-stakes problem** — blackout prevention, not a nice-to-have.
- ✓ **Matches our team's strength** — real AI depth, not just UI wiring.

Fit Score



Rated highest among all seven directions explored for mission alignment, feasibility, and how well it matches what our team can actually execute.

"We're not just showing a chart of temperature — we're giving the grid a head start."

Energy Grid Heat-Demand Forecaster • FortyGuard Hackathon'26 • Team Lead Approved